APS Grade 6 Reading Analytics — Data Decision Log

Project: Independent APS Grade 6 i-Ready Reading Achievement and Equity Analysis · **Scope:** 2022–23 and 2023–24; observed through Winter 2023–24 · **Status:** Work sample

Status key — **Verified by pipeline**: automated reconciliation result · **Implemented, pending audit**: rule in code, not yet substantiated · **Reported**: figure from project notes, audit output not attached.

ID	Area	Decision / definition	Evidence	Status	Risk / next action
D01	**Demographic grain**	One profile per Student ID + Academic Year + Student Grade; original input kept unchanged.	49,132 records → 19,831 profiles; 29,301 excess records	**Implemented, pending audit**	Audit within-key field conflicts (incl. Race NA); document retained record.
D02	**Assessment grain**	One assessment per Student ID + Academic Year + Grade + Norming Window.	49,132 → 46,135 sets; 2,838 repeated groups; 2,497 students	**Implemented, pending audit**	Repeats may be genuine retests; save discrepancy audit (median 20 pts; 399 groups ≥50 pts).
D03	**Assessment selection**	Keep one valid scored record per key (`02_iready_clean.R`).	Authoritative attempt not established	**Implemented, pending audit**	Record tie-break rule; compare earliest / latest / highest-score attempts.
D04	**Merge**	Join selected assessments to demographic profiles at assessment grain.	46,135 → 46,135 merged; 0 unmatched	**Verified by pipeline**	Coverage only; check key uniqueness and field consistency.
D05	**Achievement outcome**	Mid or Above = 1; other valid placements (incl. Early On) = 0; missing = NA.	Rate = 100 × Mid/Above ÷ valid placements	**Implemented**	Document category handling; Early On is not distance from the cutoff.
D06	**Equity groups**	Mutually exclusive, Hispanic-first: Hispanic/Latino; Black non-Hispanic; Other non-Hispanic; Unknown.	`case_when()` classification	**Implemented**	Not overlapping groups; confirm against district policy; document Race missingness.
D07	**Reporting periods**	Five windows observed; Spring 2023–24 projected only.	Source-period scope confirmed	**Verified**	Label projections in every chart; Spring vs Spring = different cohorts.
D08	**Model A — transition**	Pooled 2022–23 Winter→Spring placement probabilities applied to 2023–24 Winter (File 05).	17.1% forecast; 2,207/2,207 records scored	**Reported**	Temporal stability and held-out validation unproven.
D09	**Model B — logistic**	Spring Mid/Above from prior Winter scale score (File 06).	≈17.6%, aggregated from school forecasts	**Reported**	Brier score is in-sample, not held-out validation.
D10	**School comparison**	Prior Spring vs projected Spring, in percentage points, with assessed count.	Some locations have three assessed students	**Implemented**	Different cohorts, not causal; missing priors stay NA.
D11	**Small cells**	Flag small school and school × equity groups for disclosure review.	2 review flags	**Reported**	Apply district-approved suppression before public release.
D12	**QA and parsing**	Automated reconciliation and range checks.	0 QA failures; CSV parser warnings unresolved	**Verified by pipeline**	Run `readr::problems()` after each `read_csv()`.

Open items: (1) investigate readr parser warnings — D12; (2) complete demographic field-conflict audit — D01; (3) compare attempt-selection rules — D02–D03; (4) validate both forecasts on held-out matched students against a constant-probability baseline — D08–D09.

Disclosure: student-level data and unsuppressed small cells must not be published to Tableau Public, GitHub or public slide files without authorization. Update a status only when its audit output is saved and reviewed.